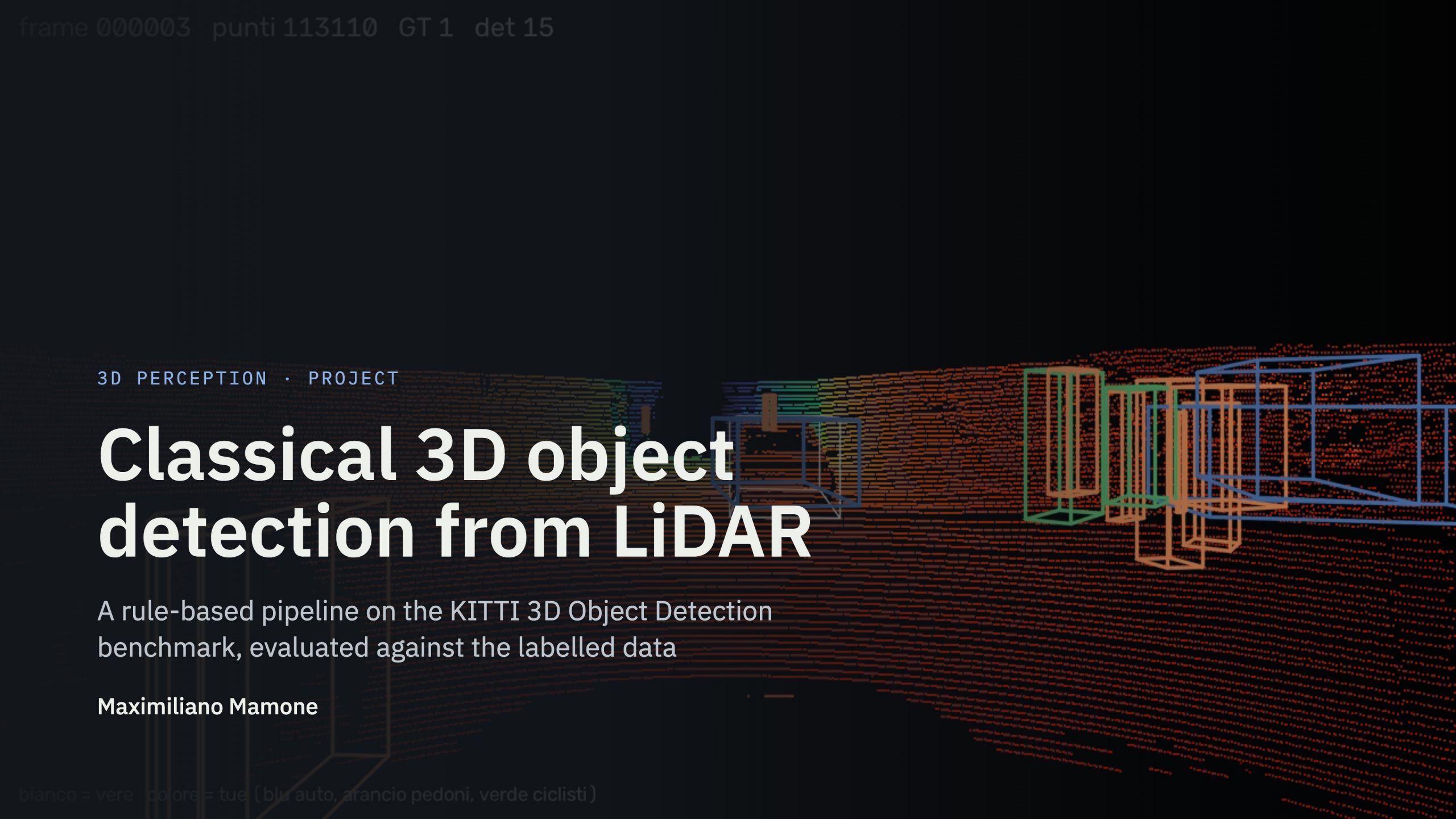


3D PERCEPTION · PROJECT

Classical 3D object detection from LiDAR

A rule-based pipeline on the KITTI 3D Object Detection benchmark, evaluated against the labelled data

Maximiliano Mamone

OBJECTIVE

How far can pure geometry go?

01

Detect

Cars, Pedestrians and Cyclists as oriented 3D boxes, from a single LiDAR scan.

02

Without learning

Only geometry and hand-set thresholds: RANSAC, a BEV grid, connected components, size rules.

03

Measure

Compare with the KITTI labels using the official 3D IoU thresholds, and see where the method stops.

DATA

KITTI 3D Object Detection

7,481

labelled training frames

1.7 m

Velodyne HDL-64E height above the road, spinning at ~10 Hz

~115k

points per frame (110–120k), range up to 120 m

4

calibrated cameras; labels refer to camera 2 (colour, left)

⚠ Labels are in **rectified camera-0** coordinates, the points in the **LiDAR** frame: a transformation is needed before any comparison.

METHOD

Pipeline overview

01
Ground removal
RANSAC plane, τ = 0.2 m

02
BEV grid
log-polar, 200 × 1000 cells

03
Clustering
8-connected cells + 3×3 closing

04
Box fitting
edge-fit over 180 orientations

05
Classify
size rules on L × W × H

06
Evaluation
3D IoU against KITTI labels

INPUT

One Velodyne scan: ~115k points (x, y, z, intensity)

OUTPUT

Oriented 3D boxes (x, y, z, l, w, h, yaw) with a class

METHOD · 01

Ground removal with RANSAC

1. Pick 3 random points among the low ones (z < −1.5 m, close to the road)

2. Build the plane through them and let every point vote if it lies within τ

3. Repeat 500 times, keep the plane with the most inliers, drop them

τ = 0.1, 0.15 and 0.2 m tried: 0.2 m looked best. The plane is kept to place box bottoms and measure heights later.

INLIER TEST

$$|\mathbf{n} \cdot \mathbf{p} + d| < \tau$$

n unit normal, pointing up ($n_z > 0$)
d plane offset
p a LiDAR point
τ = 0.2 m

METHOD · 02

Bird's-eye view on a log-polar grid

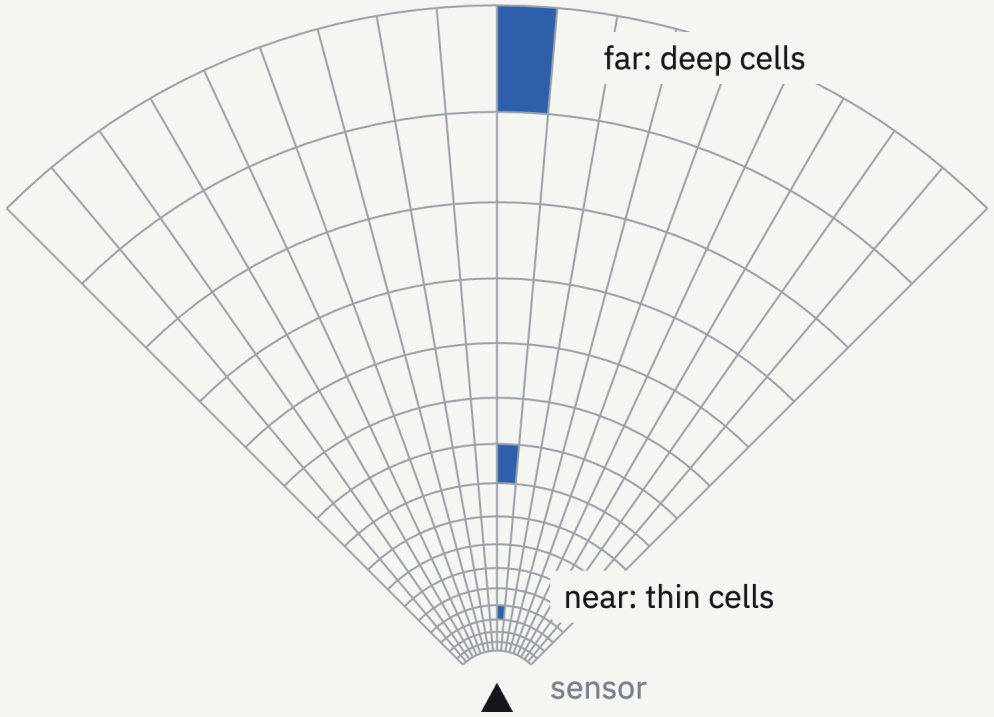
Range 2–64 m. Returns degrade past ~70 m; under 2 m is the ego car.

200 × 1000 cells: 200 rings, 1000 angular sectors (0.36° each).

```
i_r = ⌊ N_R · ln(r / r_min) / ln(r_max / r_min) ⌋
```

Ring depth grows with range, like the laser ring spacing: 0.09 m at 5 m, 0.35 m at 20 m, 0.87 m at 50 m.

A linear grid would lose point density per cell with distance and split far objects.



METHOD · 03

Clustering occupied cells

Occupancy: a cell is on if at least one point falls in it.

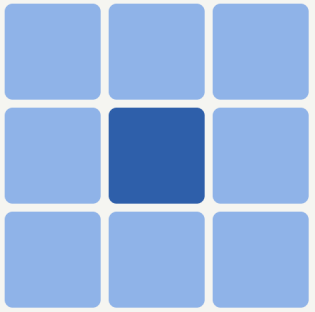
Morphological closing (3 × 3) fills one-cell gaps between parts of the same object.

Connected components with king adjacency: diagonal neighbours count too.

Back to the points: each cluster's points are taken from the original cloud; clusters under 5 points are dropped.



4-adjacency



King (8) adjacency, used

METHOD · 04

Fitting an oriented box

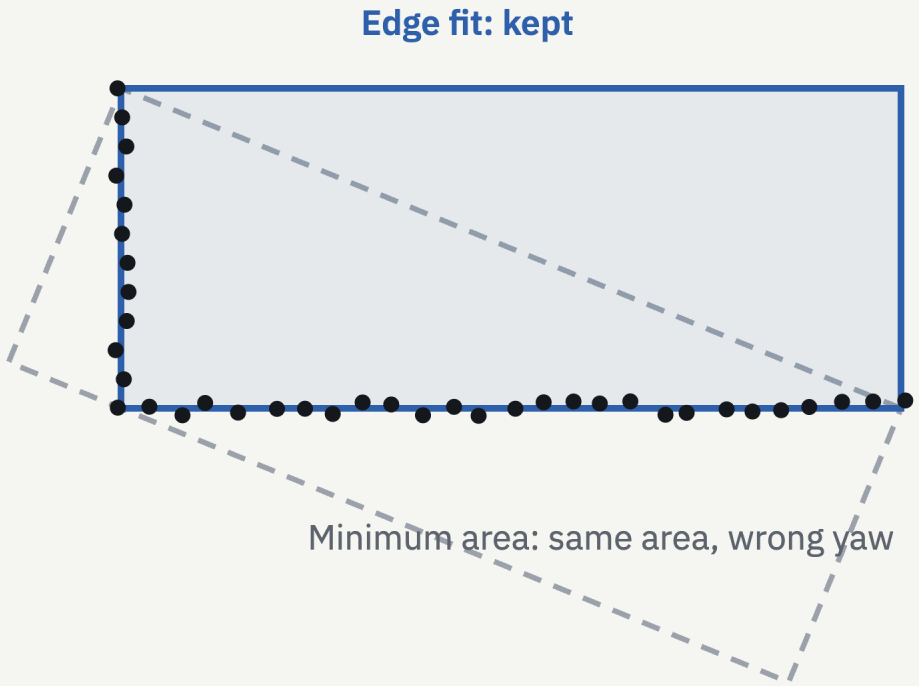
Sweep 180 orientations, 0° to 90° in 0.5° steps: a rectangle repeats every 90°.

Project the points on u and v ⊥ u; min/max extents give the rectangle.

Keep the box whose edges are closest to the points (mean distance to the nearest edge), not the minimum-area one.

Height from the RANSAC plane under the centre to the highest point.

A car seen from a corner is an L: minimum area is ambiguous between the true box and one aligned with the diagonal.



METHOD · 05

Classification by exclusion on L × W × H

#	Condition (metres)	Result
1	h < 1.0 or h > 2.5 or l > 6.0 or w > 2.5	rejected: curbs, walls, trees
2	w > 1.0 or l > 2.2	Car
3	h < 1.2	rejected: bushes, bins, posts
4	l > 1.2	Cyclist
5	otherwise	Pedestrian

Car completion

The LiDAR sees only the near side: Car boxes shorter or narrower than the KITTI mean (3.9 × 1.6 m) are extended away from the sensor.

Camera field of view

Only boxes whose centre projects inside image 2 are kept.

EVALUATION

Comparing with the KITTI labels

Same frame

Labels go from rectified camera-0 to LiDAR with the inverse of the calibration chain:

$(R0_rect \cdot Tr_velo_to_cam)^{-1}$

3D IoU does not change under a rigid transformation. P2 (camera 2) is used only to keep detections inside the image: only objects visible there are labelled.

Official protocol

Greedy matching by decreasing 3D IoU, thresholds Car 0.7, Pedestrian 0.5, Cyclist 0.5.

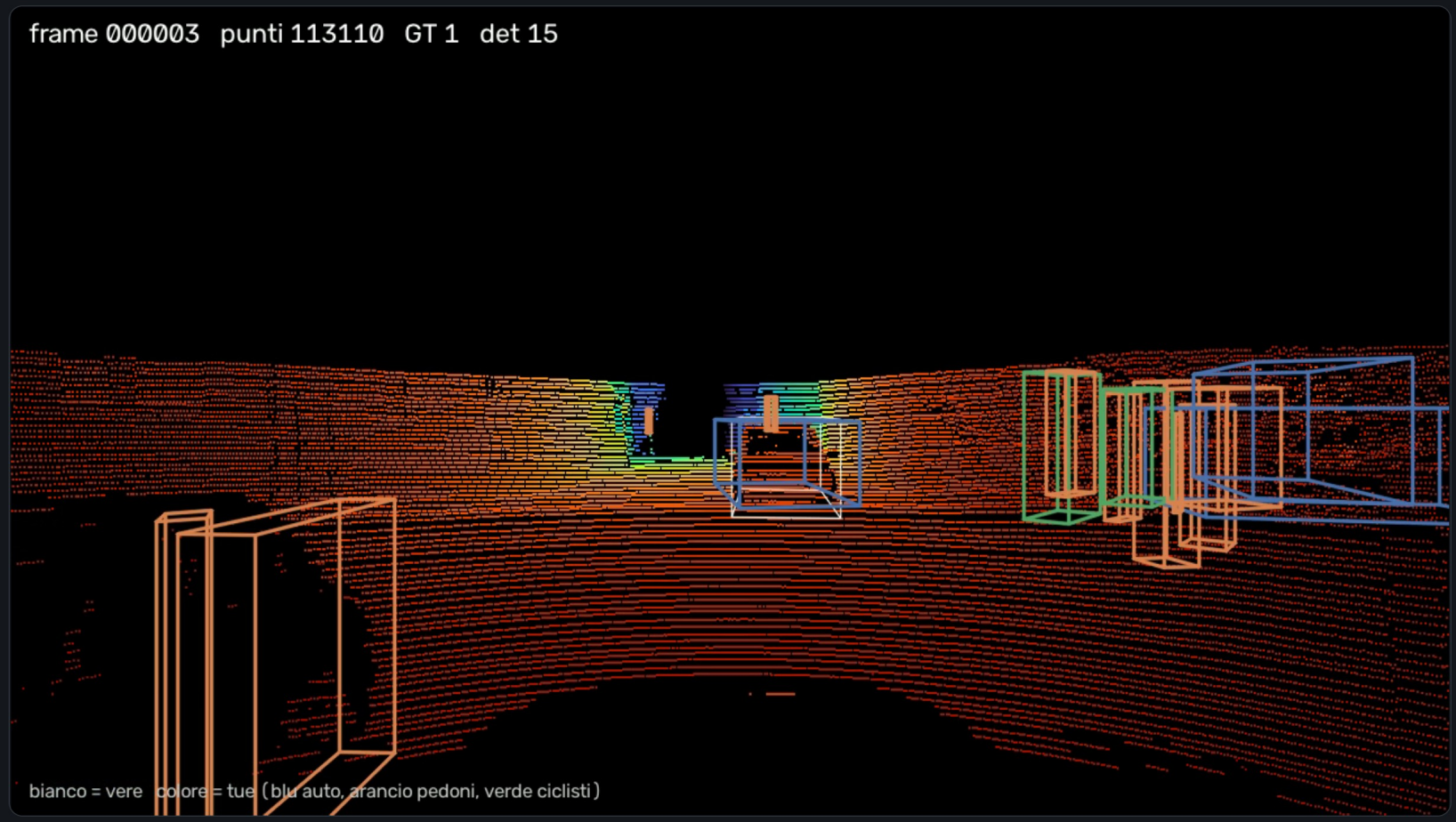
Moderate subset: occlusion ≤ 1, truncation ≤ 0.3, 2D height ≥ 25 px.

Ignored (not TP, not FN, a detection on them is not FP): Van, Person_sitting, objects outside Moderate.

DontCare not handled: precision slightly underestimated.

RESULTS · QUALITATIVE

The sensor's point of view



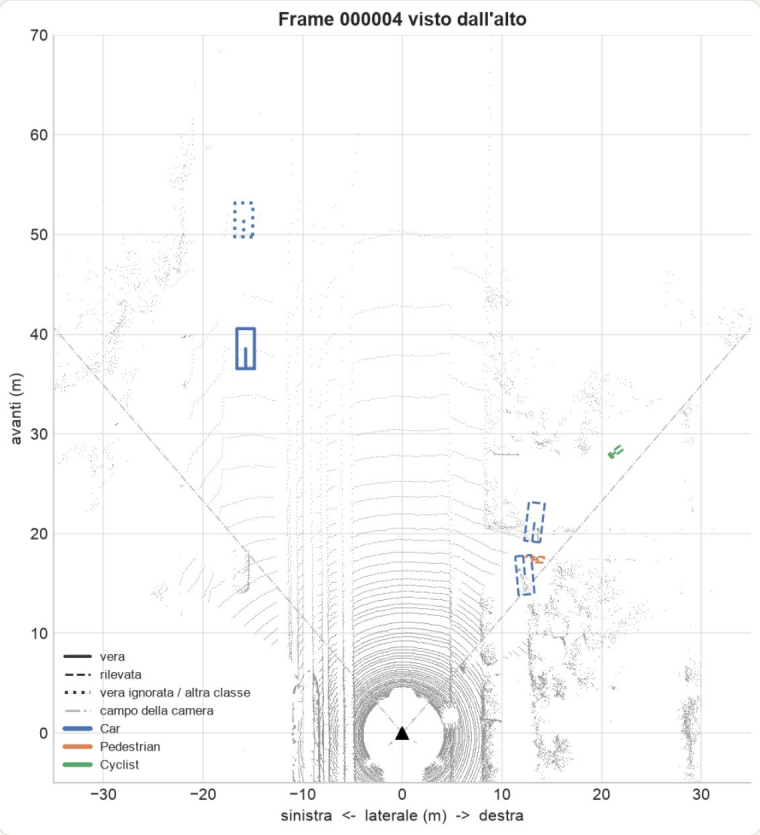
- Ground truth
- Car
- Pedestrian
- Cyclist

Frames 0–79 at 2 fps. Each frame is a different scene, not a sequence.

Top bar: frame id, points, labels and detections.

RESULTS · QUALITATIVE

One frame from above



Reading the plot

Solid label (vera) · **dashed** detection (rilevata) · **dotted** label ignored by the protocol · **dash-dot** camera field of view. Colour is the class.

A typical failure: 5 detections, 0 matches

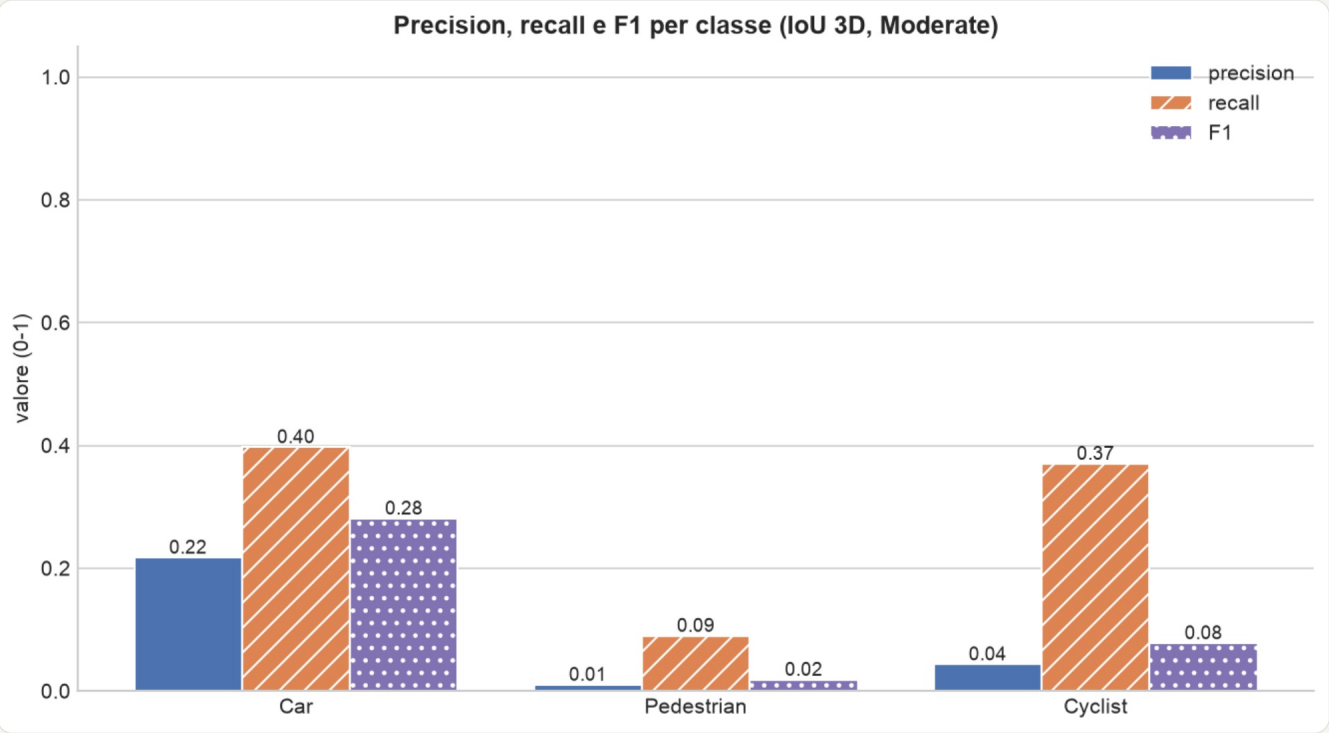
The one valid label, a car at ~38 m, gets too few returns and is missed (1 FN).

Unlabelled clutter on the right becomes 2 Car, 2 Pedestrian and 1 Cyclist box (5 FP).

The car at ~51 m (dotted) is ignored by the protocol.

RESULTS · 7,480 FRAMES · 3D IOU · MODERATE

Cars work, the small classes don't



Car
P 0.22 · R 0.40
0.28

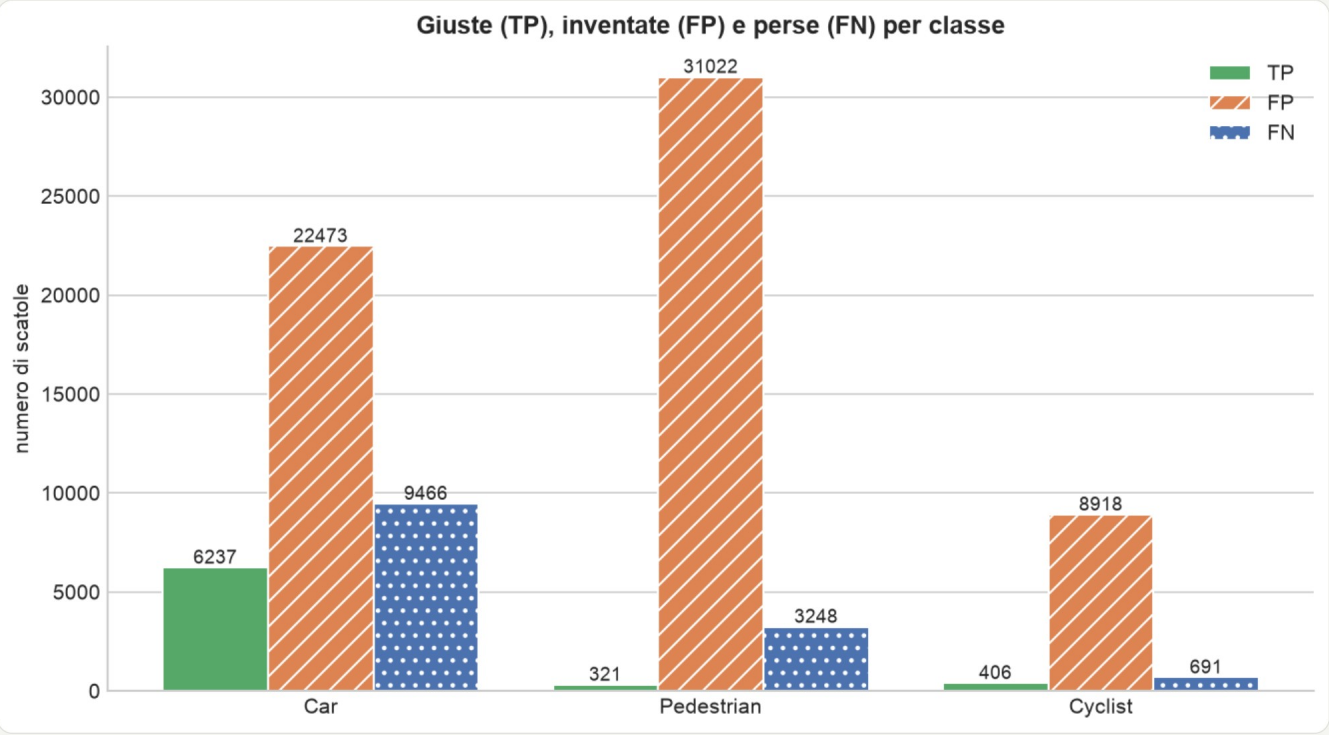
Pedestrian
P 0.01 · R 0.09
0.02

Cyclist
P 0.04 · R 0.37
0.08

Big number = F1 score

RESULTS · ERRORS

False positives dominate



62,413

false positives, against 6,964 true positives

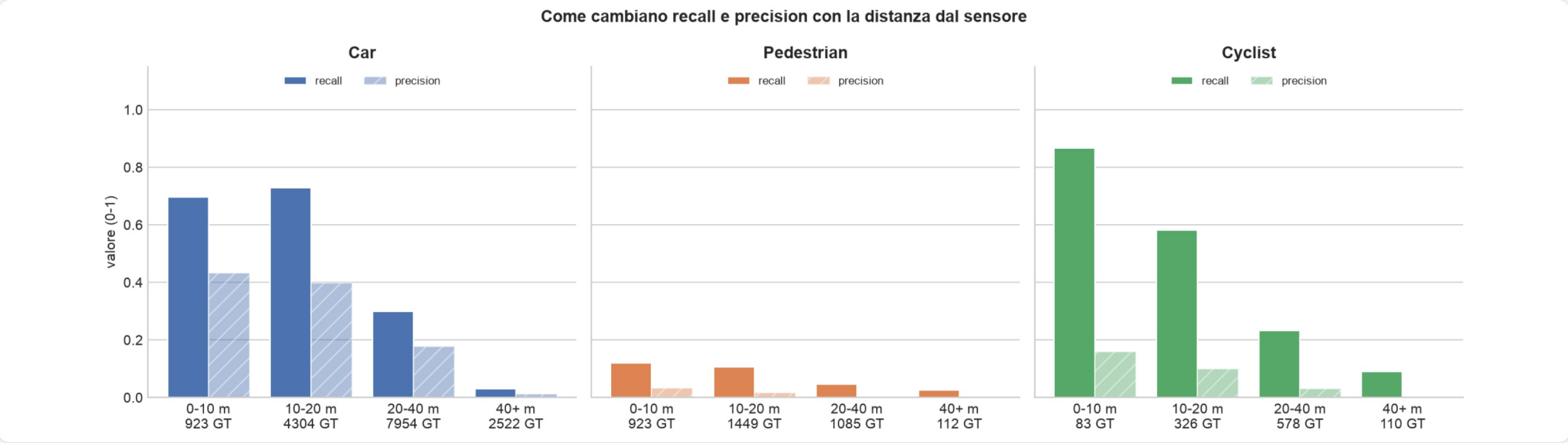
9.7 vs 2.7

detections vs valid labels per frame, on average

Half of the FPs are **Pedestrians**: poles, trunks and signs have the same size as a person.

RESULTS · BY DISTANCE

Performance falls with range



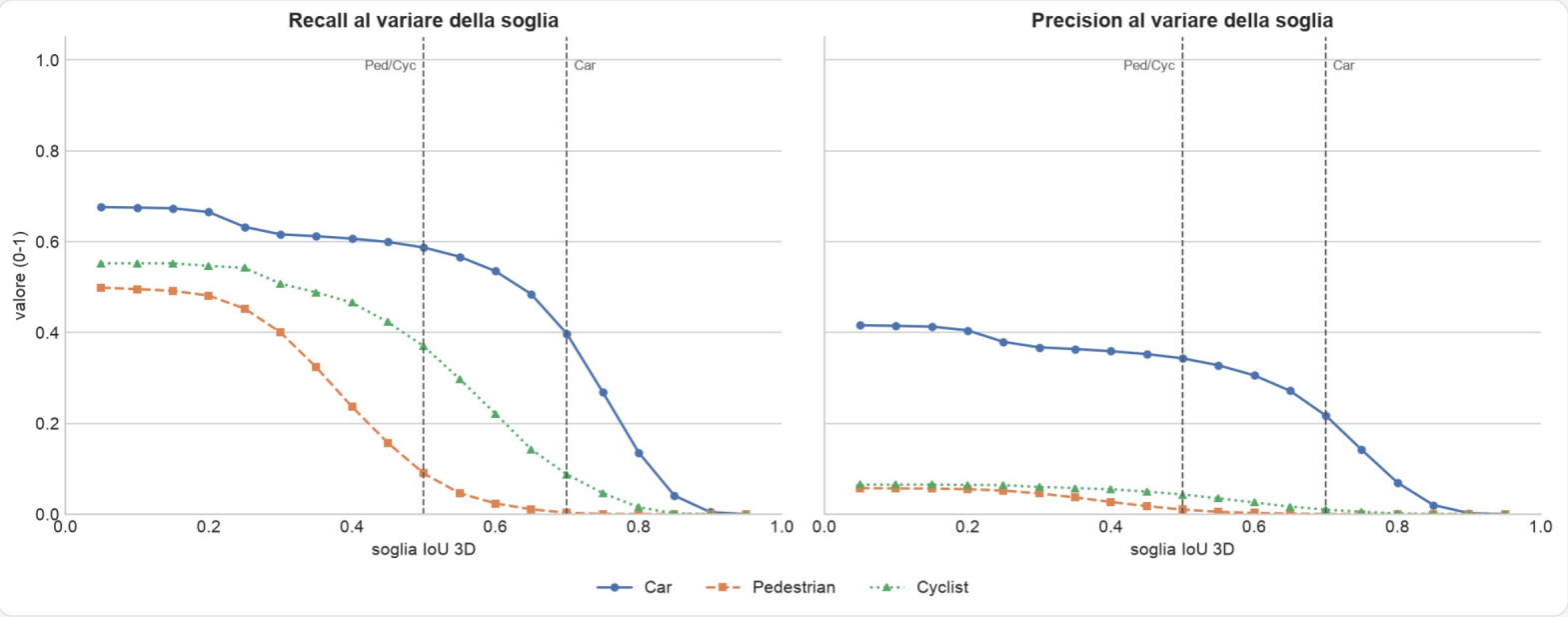
Car recall ≈ 0.7 within 20 m, 0.30 at 20–40 m, ≈ 0.03 past 40 m.

Half of all Car labels (7,954 of 15,703) sit at 20–40 m, where recall drops.

Cyclists reach 0.86 recall under 10 m; **Pedestrians** stay ≤ 0.12 everywhere.

RESULTS · IOU THRESHOLD

Found, but not precisely boxed



Car: at IoU 0.5 recall ≈ 0.59 and precision ≈ 0.34, vs 0.40 / 0.22 at 0.7.

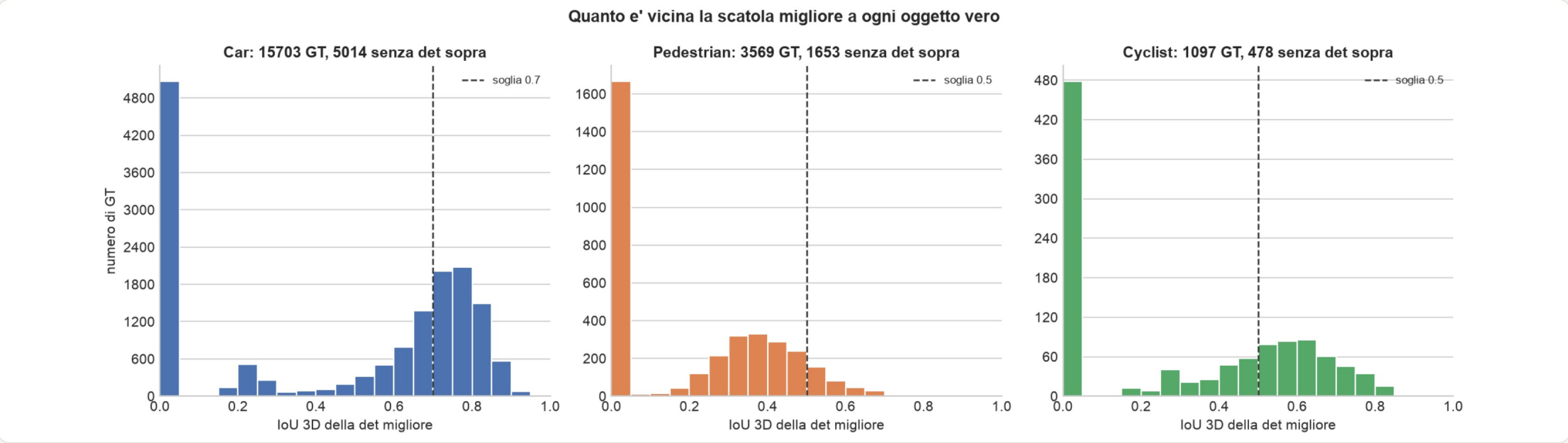
Pedestrian: recall ≈ 0.50 with loose overlap, 0.09 at 0.5.

Cyclist: ≈ 0.55 with loose overlap, 0.37 at 0.5.

Many objects are detected; the box is what fails the test.

RESULTS · LOCALISATION

Best overlap for every labelled object



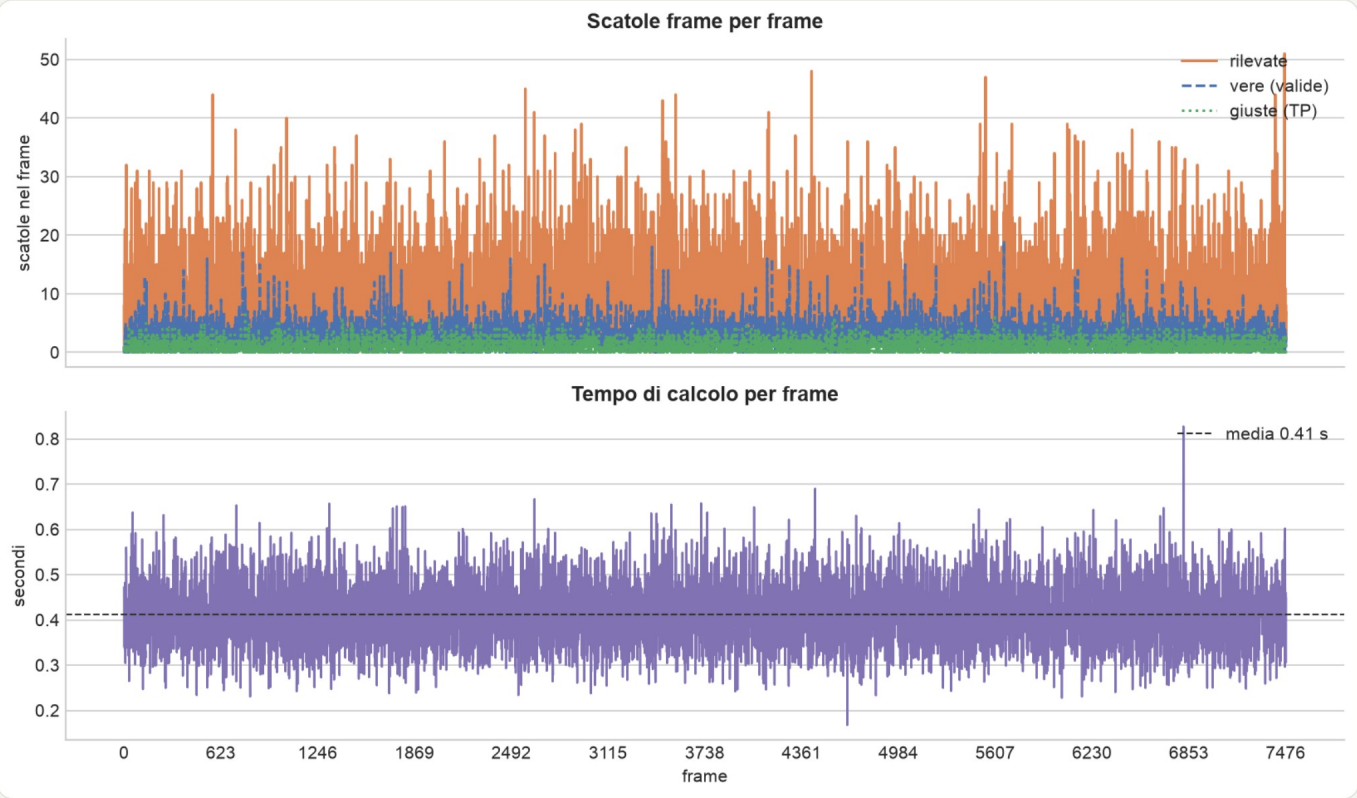
Car: 32% have no overlapping box; many sit at 0.6–0.7, just under the threshold.

Pedestrian: 46% have no box; the rest peak around 0.35.

Cyclist: 44% have no box; the rest centre on 0.5–0.65.

RESULTS · FRAME BY FRAME

Consistent across frames, ~0.4 s each



0.41 s

mean time per frame (0.17–0.83 s), Python and NumPy

~51 min

for the full run of 7,480 frames

Detections (orange) stay well above labels (blue) in every part of the dataset: the FP problem is systematic, not a few bad scenes.

CONCLUSIONS

Where geometry stops

Works

Near-range cars: recall ≈ 0.7 within 20 m

Every decision is a readable rule

No training data, ~0.4 s per frame

Breaks

Precision: size alone can't tell a person from a pole

Range: sparse returns past 20 m

Box accuracy on small objects

Next

Shape and intensity features to classify

Handle DontCare regions

Compare with a learned detector on the same protocol

THANK YOU

Questions?

Maximiliano Mamone